

Kevin Kattakayam

Machine Learning & Computer Vision | IEEE-Published | Internship & Full-Time | Graduating 2027

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EDUCATION

Karunya Institute of Technology and Sciences

2023 – 2027 (Expected)

B.Tech, Computer Science and Engineering (AI & ML) – CGPA: 8.39/10

Coimbatore, India

Viswajyothi CMI Public School, Angamaly – Secondary, CBSE (2021): 92.2%; Higher Secondary, CBSE (2023): 86%

PUBLICATIONS & ACHIEVEMENTS

- **IEEE Publication (2026):** “Early Lung Cancer and Heart Disease Prediction from Clubbed Fingers Using Machine Vision”.
ieeexplore.ieee.org/document/11469562
- **Completed Amazon ML Summer School 2026** (Amazon)

RESEARCH & ENGINEERING PROJECTS

Schamroth Window Detection – Digital Clubbing Screening

github.com/KevinKattakayam/schamroth

Python, MediaPipe, OpenCV, CLAHE, Streamlit (basis of the IEEE publication)

- Built a screening pipeline detecting digital clubbing from the Schamroth window test: MediaPipe hand-landmark localization of opposing index fingertips with a center-based fallback for close-up frames.
- Designed the gap-analysis algorithm around light transmission, deriving a gap-presence score from bright/dark pixel ratios, contrast, and left-right symmetry, and mapping it to 5 clinical risk levels with quality-weighted confidence scoring.
- Built an evaluation harness reporting accuracy, sensitivity, specificity, F1, and a confusion matrix over a 65-image test set spanning healthy, simulated, and clinical clubbing cases.

V-JEPA Video Representation Learning

github.com/KevinKattakayam/finetunevjpea

PyTorch, V-JEPA, CUDA

- Adapted pretrained V-JEPA self-supervised representations for video classification using attentive probes on UCF101.
- Ran experiments over temporal sampling strategies, probe configurations, and batch sizing under constrained GPU memory.

PharmaTrace – Multi-Agent Clinical Verification

github.com/KevinKattakayam/pharma_trace

Python, FastAPI, LangGraph, Groq/Llama 3.3 70B, Whisper, Tesseract, PostGIS

- Built a 6-agent LangGraph pipeline over OpenFDA, RxNav, and CDSCO registries, computing FAERS signal strength and severity-sorted interaction warnings rather than naive report counts.
- Constrained LLM outputs with strict Pydantic schemas and a field-validator “jargon firewall” to structurally block hallucinated medical terminology; logged every verification to a SHA-256 hash-chained audit chain.

EXPERIENCE

TheAiSignal – Full Stack AI Developer Intern

May 2026 – Jul 2026

- Engineered AI-powered features across a React frontend, FastAPI backend, and REST API layer in production.

Karunya Computer Technology Centre – Generative AI Trainee

Dec 2024 – Present

- Build LLM-powered applications with a team across multiple generative AI projects using LangChain, ChromaDB, and FAISS.
- Apply retrieval-augmented generation, prompt engineering, and OCR-based extraction to document processing workflows.

YoungMinds Technology Solutions Pvt Ltd – Generative AI Intern

May 2025 – Jul 2025

- Developed generative AI applications and automation tools in Python against LLM and REST APIs.

Karunya Innovation and Design Studio (KIDS) – Generative AI Intern

May 2025 – Jun 2025

- Built a 5-agent CrewAI pipeline for automated patent drafting, prior-art search, and compliance checking.

Octanet Services Pvt Ltd – Web Development Intern

May 2024 – Jun 2024

- Built responsive web applications using HTML, CSS, and JavaScript following mobile-first design principles.

TECHNICAL SKILLS

ML & CV: PyTorch, Self-Supervised Learning (V-JEPA), Attentive Probing, Transfer Learning, MediaPipe, OpenCV, Evaluation Design

LLM & NLP: LangGraph, LangChain, CrewAI, Hugging Face, LLM APIs, RAG, Prompt Engineering, Structured Outputs, Sentence Transformers

Languages & Tools: Python, SQL, JavaScript, TypeScript, Java, NumPy, Pandas, ChromaDB, FAISS, PostgreSQL, Jupyter, CUDA, Git

CERTIFICATIONS

Azure AI Engineer Associate, AI-102 (Microsoft) · SQL AI Developer Associate, DP-800 (Microsoft) · Oracle Agentic AI Foundations Associate · Oracle OCI AI Foundations Associate